Math 3160H: Assignment II

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March 25, 2026

Problem 1. A simple pendulum initially has the length ℓ_0 and makes an angle θ_0 with the vertical line. It is then released from this position. If the length of the pendulum increases with time t according to $\ell = \ell_0 + \epsilon t$, where ϵ is a small constant, then the angle θ that the pendulum makes with the vertical line, assuming that the oscillations are small, is the solution of the initial value problem

$$(\ell_0 + \epsilon t)\frac{d^2\theta}{dt^2} + 2\epsilon\frac{d\theta}{dt} + g\theta = 0, \quad \theta(0) = \theta_0, \quad \frac{d\theta}{dt}(0) = 0$$

where g is the acceleration due to gravity. Find the solution of the above initial value problem.

Solution 1. Consider the change of variables

$$\ell - \ell_0 = \epsilon t.$$

This gives us $d\ell = \epsilon dt$ and so

$$\frac{d\theta}{dt} = \epsilon\frac{d\theta}{d\ell}$$

and

$$\epsilon^2\frac{d^2\theta}{d\ell^2}$$

so the ODE becomes

$$\ell\epsilon^2\frac{d^2\theta}{d\ell^2} + 2\epsilon^2\frac{d\theta}{d\ell} + g\theta = 0$$

which we can divide through by ϵ^2 to see that

$$\ell\frac{d^2\theta}{d\ell^2} + 2\frac{d\theta}{d\ell} + \frac{g}{\epsilon^2}\theta = 0$$

which is very nearly the Bessel equation, we just need to fiddle a bit more. We can multiply by ℓ to get

$$\ell^2\frac{d^2\theta}{d\ell^2} + 2\ell\frac{d\theta}{d\ell} + \frac{g\ell}{\epsilon^2}\theta = 0$$

which is of a form we know,

$$x^2y'' + (1-2a)xy' + \left[b^2c^2x^{2c} + (a^2 - p^2c^2)\right]y = 0$$

which has solution

$$y = c_1x^aJ_p(bx^c) + c_2x^aY_p(bx^c).$$

Here then

$$\begin{aligned} 1 - 2a &= 2 \\ b^2c^2 &= g/\epsilon^2 \\ 2c &= 1 \\ a^2 - pc^2 &= 0 \end{aligned}$$

which tells us that

$$\begin{aligned} c &= 1/2 \\ a &= -1/2 \\ b &= 2\sqrt{g}/\epsilon \\ p &= 1 \end{aligned}$$

so we get a solution of the form

$$\theta(\ell) = c_1 \ell^{-1/2} J_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right) + c_2 \ell^{-1/2} Y_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right).$$

Now recall we originally had initial conditions in t so we need to change those to ℓ. Since $\ell = \ell_0 + \epsilon t$, when $t = 0$ $\ell = \ell_0$ so $\theta(t=0) = \theta_0 \implies \theta(\ell = \ell_0) = \theta_0$ and $\frac{d\theta}{dt}(t=0) = 0 \implies \frac{d\theta}{d\ell}(\ell = \ell_0) = 0$. So we take the derivative of our solution,

$$\begin{aligned}
\frac{d\theta(\ell)}{d\ell} &= \frac{d}{d\ell}\left[c_1 \ell^{-1/2} J_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right) + c_2 \ell^{-1/2} Y_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right)\right] \\
&= -\frac{1}{2\ell^{3/2}}\left[c_1 J_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right) + c_2 Y_1\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right)\right] \\
&\quad + \frac{\sqrt{g}}{2\epsilon\ell}\left[c_1\left[J_0\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right) - J_2\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right)\right] + c_2\left[Y_0\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right) - Y_2\left(\frac{2\sqrt{g}}{\epsilon}\ell^{1/2}\right)\right]\right]
\end{aligned}$$

and evaluate the condition,

$$\begin{aligned}
\theta_\ell(\ell_0) = 0 &= -\frac{1}{2\ell_0^{3/2}}\left[c_1 J_1\left(\frac{2\sqrt{g\ell_0}}{\epsilon}\right) + c_2 Y_1\left(\frac{2\sqrt{g}}{\epsilon}\right)\right] \\
&\quad + \frac{\sqrt{g}}{2\epsilon\ell_0}\left[c_1\left[J_0\left(\frac{2\sqrt{g\ell_0}}{\epsilon}\right) - J_2\left(\frac{2\sqrt{g\ell_0}}{\epsilon}\right)\right] + c_2\left[Y_0\left(\frac{2\sqrt{g\ell_0}}{\epsilon}\right) - Y_2\left(\frac{2\sqrt{g\ell_0}}{\epsilon}\right)\right]\right] \\
\implies c_2 &\left\{+\frac{1}{2\ell_0^{3/2}} Y_1\left(\frac{2\sqrt{g}}{\epsilon}\right) - \frac{\sqrt{g}}{2\epsilon\ell_0}\left[Y_0(\alpha) - Y_2(\alpha)\right]\right\} = c_1\left\{\frac{\sqrt{g}}{2\epsilon\ell_0}\left[J_0(\alpha) - J_2(\alpha)\right] - \frac{1}{2\ell_0^{3/2}} J_1(\alpha)\right\}
\end{aligned}$$

for $\alpha = \frac{2\sqrt{g\ell_0}}{\epsilon}$ Now we apply the condition $\theta(\ell = \ell_0) = \theta_0$,

$$\begin{aligned}
\theta(\ell = \ell_0) = \theta_0 &= \frac{1}{\sqrt{\ell_0}}\left[c_1 J_1(\alpha) + c_2 Y_1(\alpha)\right] \\
\implies \frac{\sqrt{\ell_0}\theta_0 - c_1 J_1(\alpha)}{Y_1(\alpha)} &= c_2.
\end{aligned}$$

Since these coefficients are so long and nasty (and do not simplify, I tried in Sage) I will leave them in this form.

Problem 2. The differential equation

$$x^2 \frac{d^2E}{dx^2} - \mu x \frac{dE}{dx} - k^2 x^\nu E = 0$$

occurs in the study of the flow of current through electrical transmission lines; here μ, ν are positive constants and E represents the potential difference (with respect to one end of the line) at a point a distance x from that end of the line. Find its general solution.

Solution 2. Recall we found that the ODE

$$x^2 y'' + (1-2a)xy' + \left[b^2c^2x^{2c} + (a^2 - p^2c^2)\right] y = 0$$

has solution

$$y = c_1 x^a J_p(bx^c) + c_2 x^a Y_p(bx^c).$$

The given problem is of this form with come of the coefficients being complex. We have that

$$\begin{aligned}
1 - 2a &= -\mu \\
b^2c^2 &= -k^2 \\
2c &= \nu \\
a^2 - p^2c^2 &= 0
\end{aligned}$$

which gives

$$
\begin{aligned}
a &= \frac{1}{2}(\mu+1) \\
b &= 2i\frac{k}{\nu} \\
c &= \frac{\nu}{2} \\
p &= \frac{(\mu+1)}{\nu}
\end{aligned}
$$

so we get a solution of the form

$$E = x^{\frac{1}{2}(\mu+1)}\left[c_1 J_{\frac{(\mu+1)}{\nu}}(2i\frac{k}{\nu}x^{\frac{1}{2}\nu}) + c_2 Y_{\frac{(\mu+1)}{\nu}}(2i\frac{k}{\nu}x^{\frac{1}{2}\nu})\right].$$

Problem 3.

(a) Show that the solution of the DE

$$\frac{d^2w}{dt^2} + t^m w = 0, \quad m > 0$$

can be expressed as

$$w(t) = t^{1/2}\left(AJ_{1/(m+2)}\left(\frac{2}{m+2}t^{(m+2)/2}\right) + BJ_{-1/(m+2)}\left(\frac{2}{m+2}t^{(m+2)/2}\right)\right)$$

(b) Express the solution of DE

$$\frac{d^2w}{dt^2} + w = 0$$

in terms of Bessel functions and deduce the relations given in Problem 9.2 Parts (i) and (ii).

Solution 3.

(a) Multiplying through by t^2 this is again

$$x^2y'' + (1-2a)xy' + \left[b^2c^2x^{2c} + (a^2 - p^2c^2)\right]y = 0$$

has solution

$$y = c_1x^aJ_p(bx^c) + c_2x^aJ_{-p}(bx^c)$$

for $p \notin \mathbb{Z}$. Here

$$
\begin{aligned}
1 - 2a &= 0 \\
b^2c^2 &= 1 \\
2c &= m+2 \\
a^2 - p^2c^2 &= 0
\end{aligned}
$$

so

$$
\begin{aligned}
a &= \frac{1}{2} \\
b &= 2/(m+2) \\
c &= (m+2)/2 \\
p &= 1/(m+2)
\end{aligned}
$$

which indeed gives us the solution given in the problem statement.

(b) The ODE

$$t^2 \frac{d^2 w}{dt^2} + t^2 w = 0$$

has solution

$$w(t) = c_1 t^{1/2} J_{1/2}(t) + c_2 t^{1/2} J_{-1/2}(t).$$

Since the question just asks me to "deduce"

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

we can note that the above ODE has solutions like $C_1 \sin(x) + C_2 \cos(x)$ and when plugging these in we see that

$$\frac{d^2 w}{dt^2} + w = -C_1 \sin(x) - C_2 \cos(x) + C_1 \sin(x) + C_2 \cos(x) = 0$$

and so indeed the relations must hold as we can multiply by $\sqrt{t}$. The $\sqrt{2/\pi}$ would have to come out from initial conditions as there is no normalization requirement to enforce that would imply it.

Problem 4. Use the definition of the Bessel functions to prove each of the following.

(a) $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$

(b) $J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(x^{-1} \sin x - \cos x\right)$

(c) $J_{-3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(-x^{-1} \cos x - \sin x\right)$

Solution 4. Recall that the series definition of the Bessel function is

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+n+1)} \left(\frac{x}{2}\right)^{2m+n}$$

(a) From the series definition

$$\begin{aligned} J_{-1/2}(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+1/2)} \left(\frac{x}{2}\right)^{2m-1/2} \\ &= \sqrt{\frac{2}{x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+1/2)} \left(\frac{x}{2}\right)^{2m} \end{aligned}$$

and $\Gamma(m+1/2) = \Gamma(m-1/2)(m-1/2) = \Gamma(m-3/2)(m-1/2)(m-3/2)\cdots = \frac{\sqrt{\pi}(2m-1)(2m-3)\ldots 3\cdot 1}{2^m}$ so

$$\begin{aligned} J_{-1/2}(x) &= \sqrt{\frac{2}{x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+1/2)} \left(\frac{x}{2}\right)^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(2m-1)(2m-3)\ldots 3\cdot 1\cdot 2^m} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(m)(m-1)(m-2)\ldots 3\cdot 2\cdot 1(2m-1)(2m-3)\ldots 3\cdot 1\cdot 2^m} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)(2m-2)(2m-4)\ldots 6\cdot 4\cdot 2(2m-1)(2m-3)\ldots 3\cdot 1} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \cos x. \end{aligned}$$

(b) From the series definition

$$
\begin{aligned}
J_{3/2}(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+5/2)} \left(\frac{x}{2}\right)^{2m+3/2} \\
&= \sum_{m=0}^{\infty} \frac{1}{2^{2m+1+1/2}} \frac{(-1)^m}{m!\Gamma(m+1/2+2)} x^{2m+1+1/2} \\
&= \sqrt{\frac{2x}{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!(2m+3)} x^{2m+1} && \text{Same process as previous to eval } \Gamma.
\end{aligned}
$$

Now I found this next step online as a general tip for working with Taylor series as I got stuck here. You can write

$$1/(2m+3) = \int_0^1 y^{2m+2} dy$$

which makes the series expansion

$$
\begin{aligned}
J_{3/2}(x) &= \sqrt{\frac{2x}{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!(2m+3)} x^{2m+1} \\
&= \sqrt{\frac{2x}{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \int_0^1 y^{2m+2} dy x^{2m+1} \\
&= \sqrt{\frac{2x}{\pi}} \int_0^1 y \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} y^{2m+1} x^{2m+1} dy \\
&= \sqrt{\frac{2x}{\pi}} \int_0^1 y \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} (xy)^{2m+1} dy \\
&= \sqrt{\frac{2x}{\pi}} \int_0^1 y \sin(xy) dy \\
&= \sqrt{\frac{2x}{\pi}} \frac{1}{x^2} \left[\sin(x) - x\cos(x)\right] \\
&= \sqrt{\frac{2}{x\pi}} \left[x^{-1}\sin(x) - \cos(x)\right]
\end{aligned}
$$

(c) From the series definition

$$
\begin{aligned}
J_{-3/2}(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m-3/2+1)} \left(\frac{x}{2}\right)^{2m-3/2} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m-1/2)} \left(\frac{x}{2}\right)^{2m-1/2-1}
\end{aligned}
$$

And I don't know where to go from here because $\Gamma(0-1/2) = \Gamma(-1/2)$ is undefined. I'll just show that the definition satisfies

$$x^2 y'' + xy' + (x^2 - (3/2)^2)y = 0$$

as this by definition has $J_{-3/2}(x)$ as a solution. Since this is just algebra we can pass it to Sage:

[1]:
```
clear_vars()
x = var('x')
y = -sqrt(2/(pi*x))*((1/x)*cos(x)+sin(x))
ODE = x^2*diff(y, x, 2)+x*diff(y,x)+(x^2-(3/2)^2)*y
print("full: ")
show(ODE)
```

```
print("simplified: ")
show(ODE.full_simplify())
```

[1]: full:

$$\frac{1}{4}\left(4\sqrt{2}\left(\frac{\cos(x)}{x}-\frac{2\sin(x)}{x^2}-\frac{2\cos(x)}{x^3}+\sin(x)\right)\sqrt{\frac{1}{\pi x}}-\frac{4\sqrt{2}\left(\frac{\sin(x)}{x}+\frac{\cos(x)}{x^2}-\cos(x)\right)}{\pi x^2\sqrt{\frac{1}{\pi x}}}-\frac{4\sqrt{2}\left(\frac{\cos(x)}{x}+\sin(x)\right)}{\pi x^3\sqrt{\frac{1}{\pi x}}}+\frac{\sqrt{2}\left(\frac{\cos(x)}{x}+\sin(x)\right)}{\pi^2x^4\left(\frac{1}{\pi x}\right)^{\frac{3}{2}}}\right)x^2-$$

$$\frac{1}{4}\sqrt{2}\left(4x^2-9\right)\left(\frac{\cos(x)}{x}+\sin(x)\right)\sqrt{\frac{1}{\pi x}}+\frac{1}{2}\left(2\sqrt{2}\left(\frac{\sin(x)}{x}+\frac{\cos(x)}{x^2}-\cos(x)\right)\sqrt{\frac{1}{\pi x}}+\frac{\sqrt{2}\left(\frac{\cos(x)}{x}+\sin(x)\right)}{\pi x^2\sqrt{\frac{1}{\pi x}}}\right)x$$

simplified:

0

Problem 5. In a study of planetary motion Bessel encountered the integral

$$y(x)=\frac{1}{\pi}\int_0^{\pi}\cos(n\theta-x\sin(\theta))d\theta$$

Show that $y(x)=J_n(x)$.

Solution 5. Recall that the generating function for J_n is

$$\exp\left(\frac{1}{2}x(h-\frac{1}{h})\right)=\sum_{n=-\infty}^{\infty}h^nJ_n(x).$$

If we take $h=e^{-i\theta}$ this becomes

$$\exp\left(\frac{x}{2}(e^{-i\theta}-e^{i\theta})\right)=\sum_{n=-\infty}^{\infty}e^{in\theta}J_n(x)$$

and

$$\frac{x}{2}(e^{-i\theta}-e^{i\theta})=-ix\sin(\theta)$$

so

$$\exp(-ix\sin\theta)=\sum_{n=-\infty}^{\infty}e^{-in\theta}J_n(x).$$

Now both to get us closer to the integral formula above (whose integrand can be expressed using complex exponentials) and to use the orthogonality of e^{imx} for $m\in\mathbb{Z}$ we multiply by $e^{im\theta}$,

$$\exp(im\theta-ix\sin\theta)=\sum_{n=-\infty}^{\infty}e^{i(m-n)\theta}J_n(x)$$

and integrate

$$\int_0^{\pi}\exp(im\theta-ix\sin\theta)d\theta=\sum_{n=-\infty}^{\infty}\int_0^{\pi}e^{i(m-n)\theta}d\theta J_n(x).$$

The integral

$$\int_0^{\pi}e^{i(m-n)\theta}d\theta$$

has two cases. When $m=n$, it is the integral of 1 and comes out to π. When $m\neq n$ we get

$$\int_0^{\pi}e^{ik\theta}d\theta=-i\frac{1}{k}\left[e^(ik\pi)-e^{-ik\pi}\right]=0$$

where $k=m-n$ for simplicity. Hence we obtain a Kronecker delta function for m and n which eliminates all but the $m=n$ term in the sum, which will be multiplied by π. This leaves us with

$$\int_0^{\pi}\exp(im\theta-ix\sin\theta)d\theta=\pi J_m(x).$$

Which is almost what we want. Since

$$\exp(im\theta-ix\sin\theta)=\cos(m\theta-x\sin\theta)+i\sin(m\theta-x\sin\theta),$$

$$\int_0^\pi \cos(m\theta - x\sin\theta) + i\sin(m\theta - x\sin\theta)d\theta = \pi J_m(x).$$

and since the right hand side is entirely real, the sin integral must vanish, leaving us with the desired result! Yay!

Problem 6 (SAGE). (a) Plot $P_n(x)$ for $n = 0..5$ on the same graph.

(b) Find the Zeros of $P_{12}(x)$.

(c) Use SAGE to solve each of the following differential equations:

(i) $(1-x^2)y'' - 2xy' + 30y = 0$;

(ii) $(1-x^2)y'' - 2xy' + \left(30 - \dfrac{16}{1-x^2}\right)y = 0$.

(d) Use SAGE to plot $J_n(x)$ for $n = 0..5$ on the same graph. Comment on the behaviour of the graph.

(e) Use SAGE to plot $Y_n(x)$ for $n = 0..5$ on the same graph. Comment on the behaviour of the graph.

(f) Use SAGE to plot $K_n(x)$ for $n = 0..5$ on the same graph. Comment on the behaviour of the graph.

(g) Use SAGE to plot $I_n(x)$ for $n = 0..5$ on the same graph. Comment on the behaviour of the graph.

(h) Find the first three Zeros of $J_{1/2}(x)$ using SAGE.

(i) Find the first three Zeros of $Y_1(x)$ using SAGE.

(j) Use SAGE to solve each of the following differential equations:

(i) $x^2y'' + xy' - (4x^2+9)y = 0$;

(ii) $x^2y'' + xy' + (4x^2-9)y = 0$;

(iii) $x^2y'' + 2xy' + (4x^2-30)y = 0$.

Solution 6.

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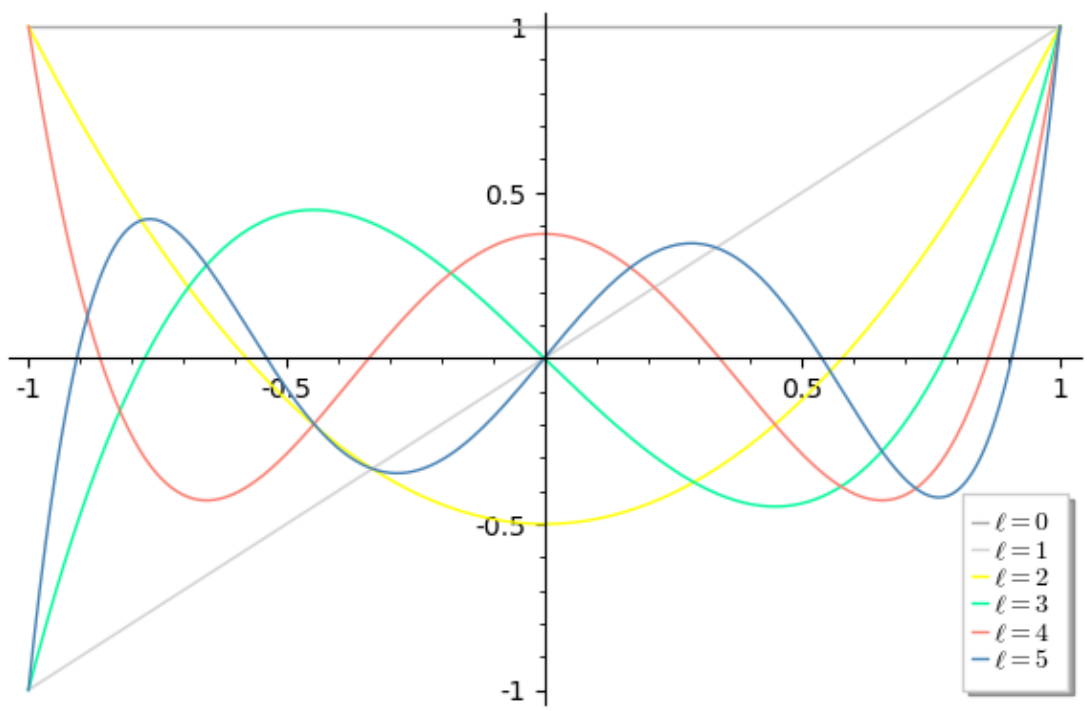
[2]:
```
import random
from IPython.core.interactiveshell import InteractiveShell
InteractiveShell.ast_node_interactivity = "all"
```

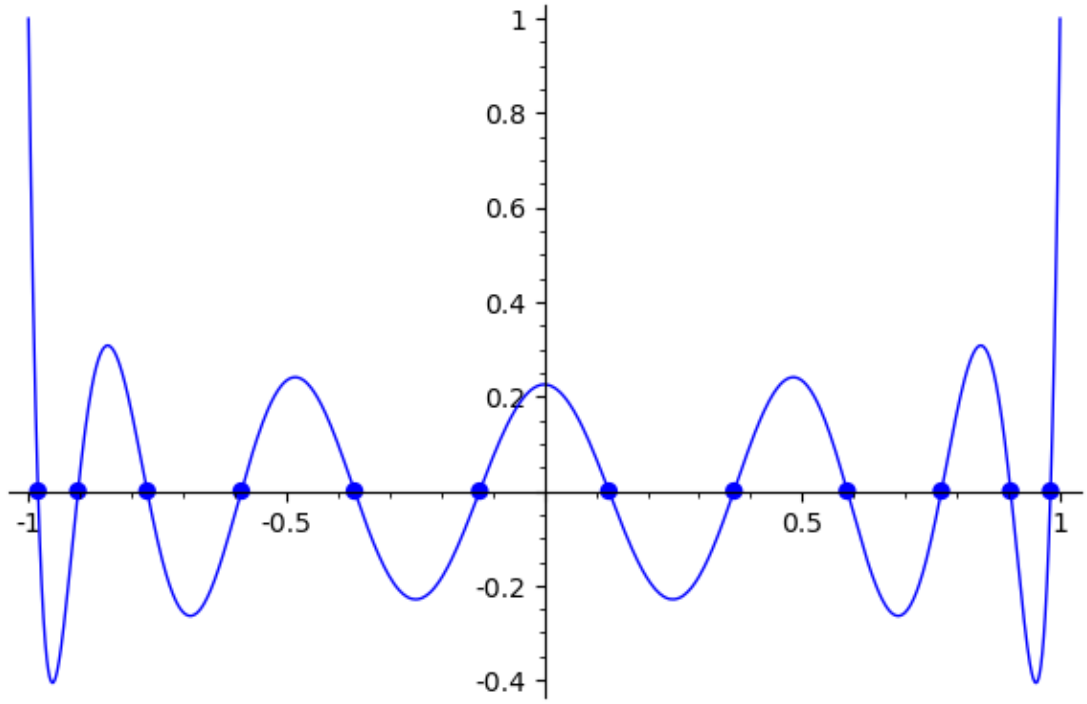
[4]:
```
# (a)
clear_vars()
x = var('x')

p = [gen_legendre_P(l, 0, x).plot((x, -1, 1),
color=random.choice(sorted(colors)),
legend_label=rf"$\ell={l}$") for l in range(0,6)]
plt = sum(p)
plt.legend(True)
show(plt)
```



[3]:

```
# (b)
clear_vars()
x = var('x')
poly = gen_legendre_P(12, 0, x)
p = plot(poly, (x,-1,1))
rts = poly.roots(ring=QQbar)
for rt in rts:
show(rt[0])
p = p + plot(point((rt[0], 0), size=45))
show(p)
```

```
-0.9815606342467193?

-0.904117256370475?

-0.769902674194305?

-0.5873179542866174?

-0.3678314989981802?

-0.1252334085114690?

0.1252334085114690?

0.3678314989981802?

0.5873179542866174?

0.769902674194305?

0.904117256370475?

0.9815606342467193?
```



[4]:
```
# (ci)
clear_vars()
x = var('x')
y = function('y')(x)
DE = (1-x^2)*diff(y, x, 2)-2*x*diff(y,x,1)+30*y==0

soln = desolve(DE, y, contrib_ode=True, show_method=True)[0][0]
show(soln.simplify_hypergeometric().simplify_full())
# Note the hypergeometrics. It seems simplify_hypergeometric does not␣
↪actually
# simplify hypergeometrics to special functions. gauss_a and gauss_b␣
↪are special cases of pFq
# hypergeomtrics and are an extension(?) of the Legendre polynomials, I␣
↪think. Getting this from Wikipedia
# and some SageMath documentation & forum threads.
```

```
y(x) == _K1*gauss_a(-5, 6, 1, -1/2*x + 1/2) + _K2*gauss_b(-5, 6, 1, -1/2*x + 1/2)
```

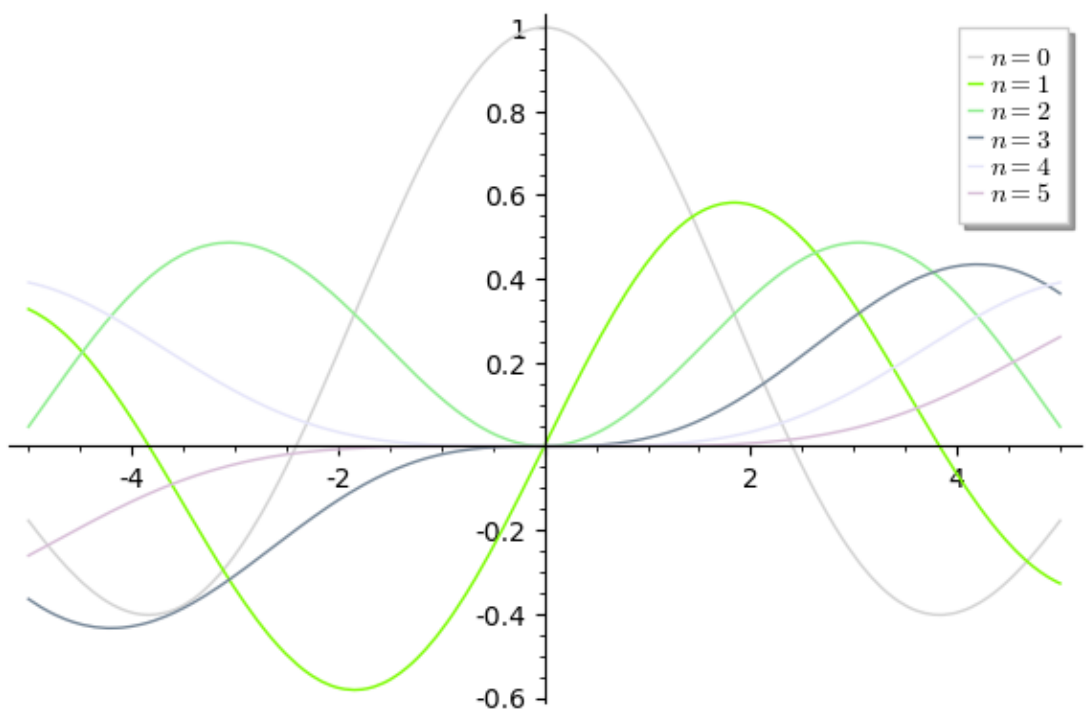
[5]:
```
# (cii)
clear_vars()
x = var('x')
y = function('y')(x)
```

```
DE = (1-x^2)*diff(y, x, 2)-2*x*diff(y,x,1)+(30-16/(1-x^2))*y==0

soln = desolve(DE, y, contrib_ode=True, show_method=True)[0][0]
show(soln)
```

```
y(x) == _K1*gauss_a(-9, 2, -3, -1/2*x + 1/2)/((x + 1)^2*(x - 1)^2) + _K2*gauss_b(-
9, 2, -3, -1/2*x + 1/2)/((x + 1)^2*(x - 1)^2)
```

[6]:

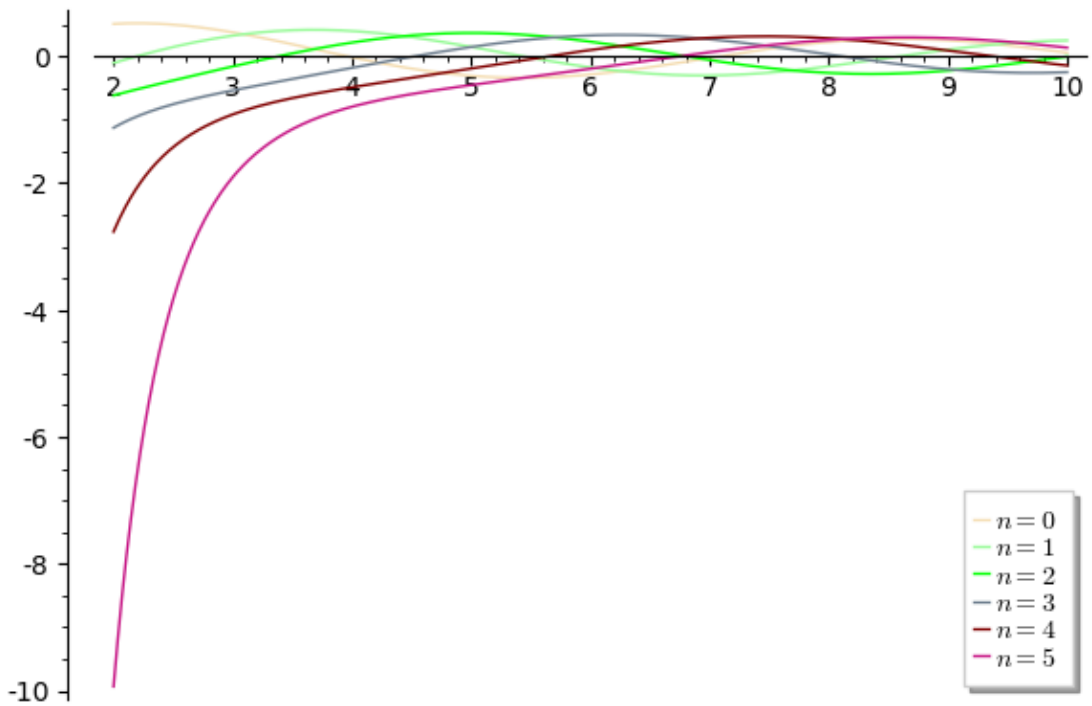
```
# (d)
clear_vars()
x = var('x')

p = [bessel_J(n, x).plot((x, -5, 5 ),
color=random.choice(sorted(colors)),
legend_label=rf"$n={n}$") for n in range(0,6)]
plt = sum(p)
plt.legend(True)
show(plt)
# We can see the expected damped oscillatory behaviour and boundedness.
```



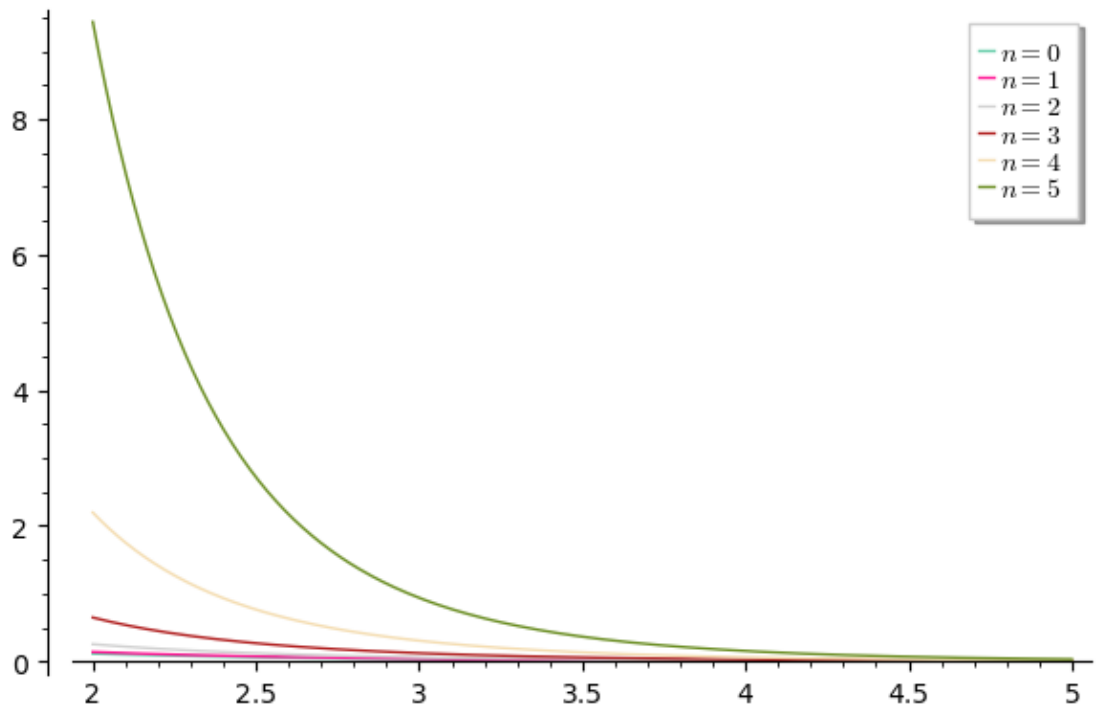
[8]:

```
# (e)
clear_vars()
x = var('x')
```

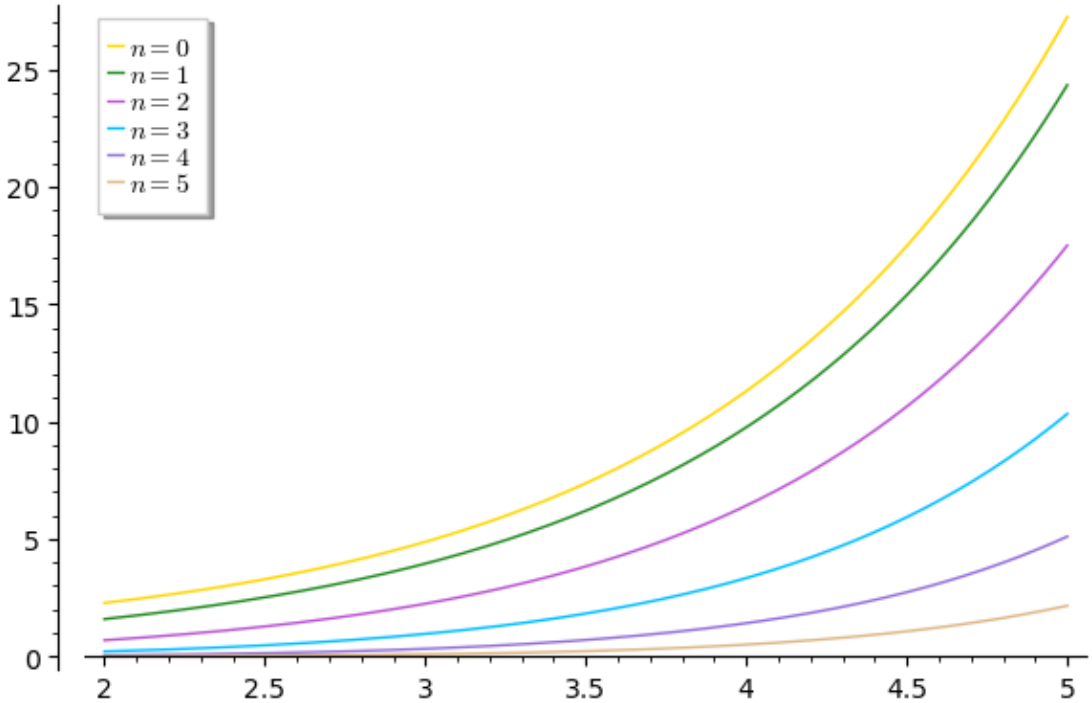
```
p = [bessel_Y(n, x).plot((x, 2, 10),
color=random.choice(sorted(colors)),
legend_label=rf"$n={n}$") for n in range(0,6)]
plt = sum(p)
plt.legend(True)
show(plt)
# We can see the damped oscillations and unboundedness near 0 requiring␣
↪its exclusion from many physical problems
```



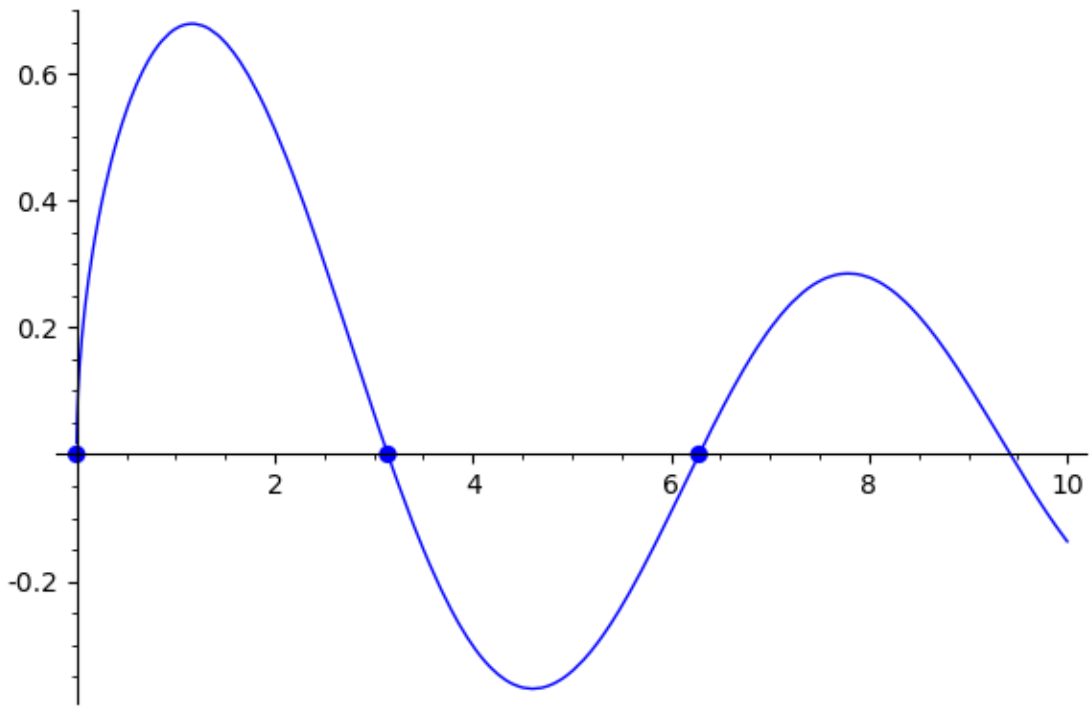
[9]:
```
# (f)
clear_vars()
x = var('x')

p = [bessel_K(n, x).plot((x, 2, 5),
color=random.choice(sorted(colors)),
legend_label=rf"$n={n}$") for n in range(0,6)]
plt = sum(p)
plt.legend(True)
show(plt)
# Again we can unboundedness near 0.
```



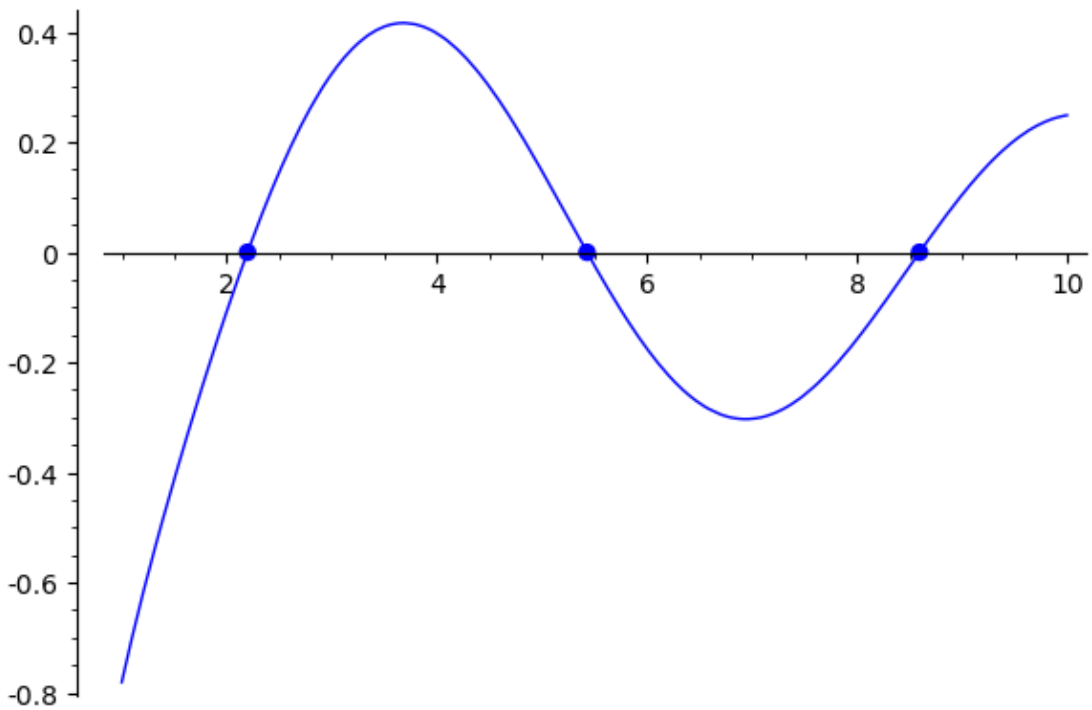
[10]:
```
# (g)
clear_vars()
x = var('x')

p = [bessel_I(n, x).plot((x, 2, 5),
color=random.choice(sorted(colors)),
legend_label=rf"$n={n}$") for n in range(0,6)]
plt = sum(p)
plt.legend(True)
show(plt)
# Here we see unboundedness for large x for a change.
```



[10]:
```
# (h)
clear_vars()
x = var('x')
poly = bessel_J(1/2, x)
p = plot(poly, (x,0,10))

for xv in [2,6]:
rt = find_root(poly, xv, xv+2)
p = p + plot(point((rt, 0), size=45))
p = p + plot(point((0, 0), size=45)) # Because there is a zero at zero␣
↪but the sage root finding tools don't play nice.
show(p)
```



[11]:
```
# (i)
clear_vars()
x = var('x')
poly = bessel_Y(1, x)
p = plot(poly, (x,1,10))

for xv in [2,5,8]:
rt = find_root(poly, xv, xv+1)
p = p + plot(point((rt, 0), size=45))
show(p)
```



[12]:
```
# (ji)
clear_vars()
x = var('x')
y = function('y')(x)
DE = (x^2)*diff(y, x, 2)+x*diff(y,x,1)-(4*x^2+9)*y==0

soln = desolve(DE, y, contrib_ode=True, show_method=True)
show(soln)
```

```
[[y(x) == I*_K1*bessel_I(3, -2*x) - _K2*bessel_Y(3, -2*I*x)], 'odelin']
```

[13]:
```
# (jii)
clear_vars()
x = var('x')
y = function('y')(x)
DE = (x^2)*diff(y, x, 2)+x*diff(y,x,1)+(4*x^2-9)*y==0

soln = desolve(DE, y, contrib_ode=True, show_method=True)
show(soln)
```

```
[[y(x) == -_K1*bessel_J(3, -2*x) - _K2*bessel_Y(3, -2*x)], 'odelin']
```

[14]:

```
# (jiii)
clear_vars()
x = var('x')
y = function('y')(x)
DE = (x^2)*diff(y, x, 2)+2*x*diff(y,x,1)+(4*x^2-30)*y==0

soln = desolve(DE, y, contrib_ode=True, show_method=True)
show(soln)
# Interestingly here Sage has opted to not go for Bessel functions␣
↪directly but instead their functional representations
# in radicals, polynomials, and trig functions. I wonder why.
```

```
[[y(x) == 1/32*(2*(16/x - 420/x^3 + 945/x^5)*cos(2*x) - 15*(16/x^2 -
    112/x^4 + 63/x^6)*sin(2*x))*_K1*sqrt(-x)*sqrt(-x/pi)/x -
    1/32*(15*(16/x^2 - 112/x^4 + 63/x^6)*cos(2*x) + 2*(16/x - 420/x^3
    + 945/x^5)*sin(2*x))*_K2*sqrt(-x)*sqrt(-x/pi)/x],
'odelin']
```

[ ]:

```
```